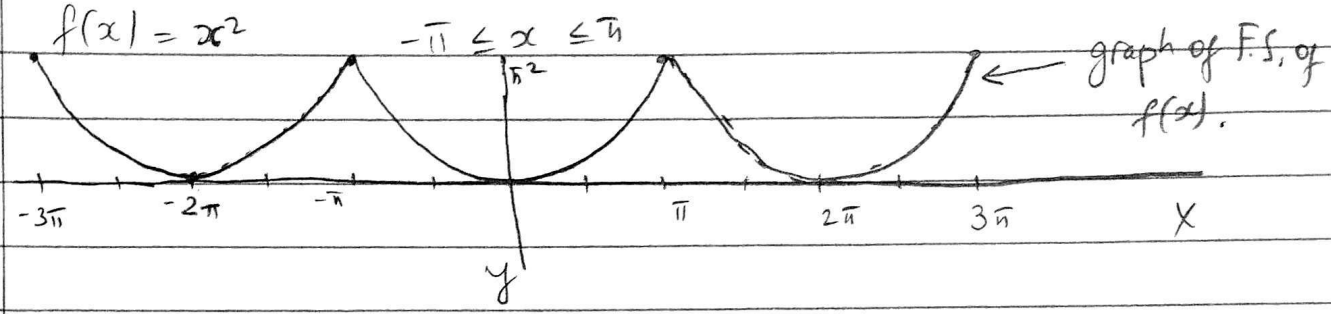
Prob. 2.

$$f(x) = \frac{f(x^+) + f(x^-)}{2}$$ for every $x \in [-\pi, \pi]$

$x = \pi$.

(1) (c)



Sol-n:

(2) $$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left[a_n \cos(nx) + b_n \sin(nx) \right]$$

because

(3) (i) $f(x)$ is piecewise smooth on $(-\pi, \pi)$ &

(ii) $f(-\pi) = f(\pi)$

We can write (2) for every $x \in [-\pi, \pi]$ since the properties in (3) hold.

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\,dx = \frac{1}{\pi} \cdot \int_{-\pi}^{\pi} x^2\,dx = \frac{2}{\pi} \cdot \int_0^{\pi} x^2\,dx =$$

$$= \frac{2}{\pi} \cdot \frac{x^3}{3} \Big|_0^{\pi} = \frac{2}{\pi} \cdot \frac{\pi^3}{3} = \frac{2\pi^2}{3}. \Rightarrow$$

(4) $$\boxed{a_0 = \frac{2\pi^2}{3}}$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx)\,dx = \frac{1}{\pi} \cdot \int_{-\pi}^{\pi} x^2 \cos(nx)\,dx =$$

$$= \frac{2}{\pi} \cdot \underbrace{\int_0^{\pi} x^2 \cos(nx)\, dx}_{I} =$$

$$I = \int_0^{\pi} x^2 \cos(nx)\, dx = \int_0^{\pi} \underbrace{x^2}_{u} \cdot d\underbrace{\left[\frac{\sin(nx)}{n}\right]}_{v} =$$

$$= x^2 \cdot \frac{\sin(nx)}{n}\Big|_0^{\pi} - \int_0^{\pi} \frac{\sin(nx)}{n}\, dx^2 =$$

$$= 0 - \frac{2}{n}\int_0^{\pi} x \sin(nx)\, dx = -\frac{2}{n}\int_0^{\pi} \underbrace{x}_{u}\, d\underbrace{\left[-\frac{\cos(nx)}{n}\right]}_{v} =$$

$$= -\frac{2}{n}\left\{ -\frac{x \cdot \cos(nx)}{n}\Big|_0^{\pi} - \int_0^{\pi} -\frac{\cos(nx)}{n}\, dx \right\} =$$

$$= -\frac{2}{n}\left\{ -\left[\frac{\pi \cdot \cos(\pi n)}{n}\right] + \frac{1}{n}\, \frac{\sin(nx)}{n}\Big|_0^{\pi} \right\} =$$

$$= \frac{2\pi \cos(\pi n)}{n^2}$$

$$a_n = \frac{2}{\pi} \cdot I = \frac{2}{\cancel{\pi}} \cdot \frac{2\cancel{\pi} \cos(\pi n)}{n} = \frac{4}{n} \cos(\pi n).$$

(5) $\boxed{a_n = \frac{4\cos(\pi n)}{n}}$ for $n = 1, 2, \ldots$

$$b_n = \frac{1}{\pi} \cdot \int_{-\pi}^{\pi} f(x) \cdot \sin(nx)\,dx = \frac{1}{\pi} \cdot \int_{-\pi}^{\pi} \underbrace{x^2 \cdot \sin(nx)}_{\text{odd}}\,dx = 0.$$

(6) $\boxed{b_n = 0}$ for $n = 1, 2, \ldots$

Then (2) becomes,

(7) $$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \left[\frac{4\cos(\pi n)}{n} \cos(nx) \right]$$